ZUKUNFT LAND

Hessian Low Mountain Range

Full data report

Generated 2026-11-04

Hesse

NUTS-3 regions: DE722, DE723, DE725

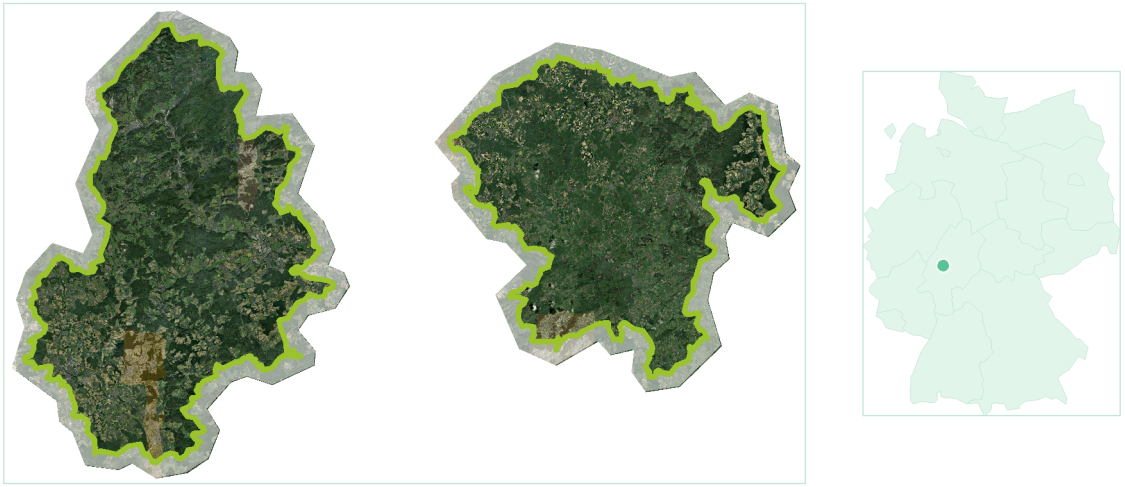


Figure 1: The Hessian Low Mountain Range Living Lab: study region (left) and its location within Germany (right). Basemap: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Agriculture

Key figures

CROPLAND AREA	NUMBER OF FARMS	AVERAGE FARM SIZE	SHARE OF ORGANIC FARMING
113,215 ha	3,832 count	54.32 ha	23.1 %

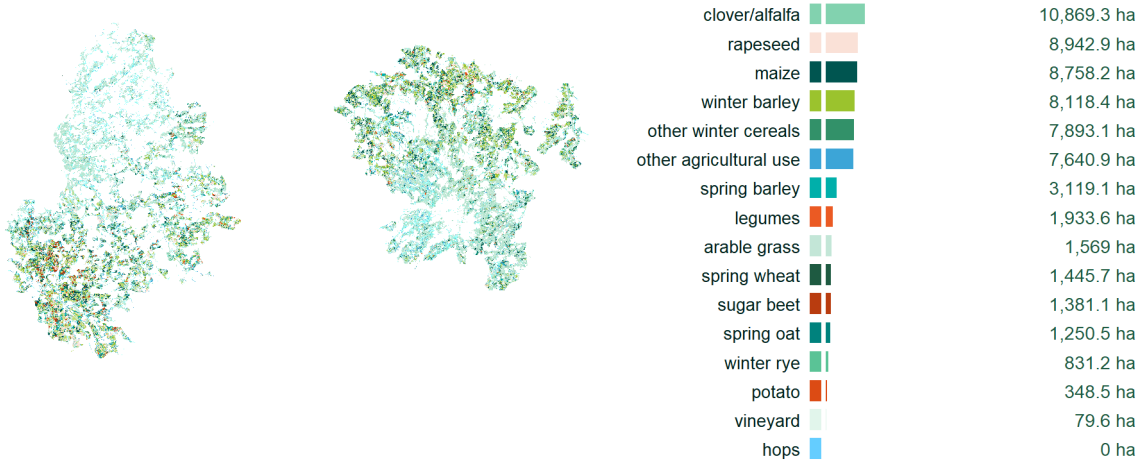


Figure 2: Map showing Agriculture in the Hessian Low Mountain Range Living Lab (Data: Crop types (DLR, 2024))

Overview

Comparatively high livestock density. High proportion of grassland. High proportion of forest. Relatively small farm sizes and field sizes

Challenges

Feed production problematic (soil unsuitable for cultivation), therefore predominantly grassland farming. Decline in dairy farming (-6% dairy cows, −14% farms). Competition for land with energy production (wind power, photovoltaics)

Climate

Key figures

GROWING DEGREE DAYS, ANNUAL SUM (BASE 5 °C)	MEAN ANNUAL TEMPERATURE	ANNUAL PRECIPITATION	PRECIPITATION OF WARMEST QUARTER
1,772 °C·d +1,085 °C·d by 2071-2100	**8.6** °C +3.9 °C by 2071-2100	**890** mm +2.8 % by 2071-2100	**226** mm -8.6 % by 2071-2100

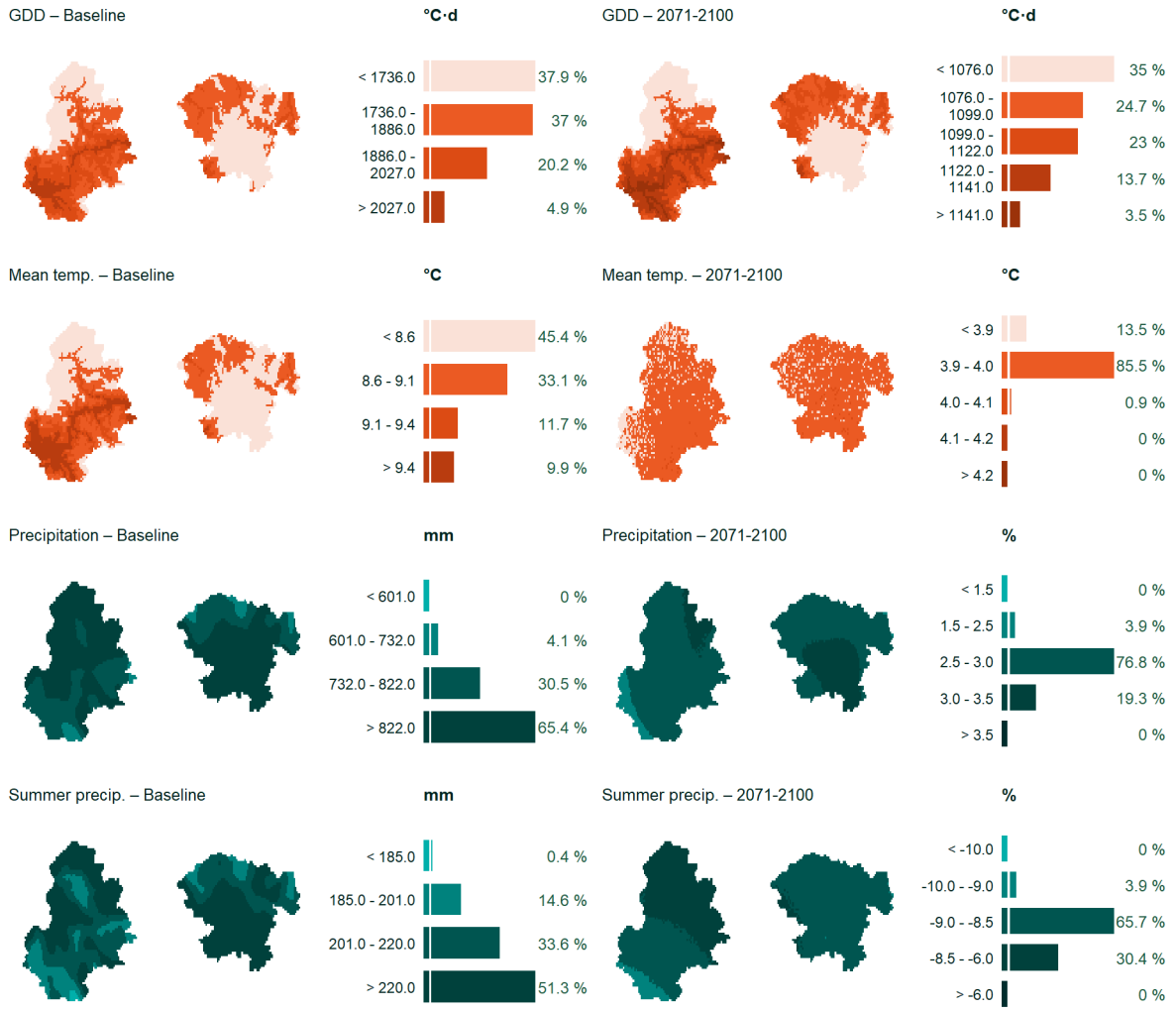


Figure 3: Map showing Climate in the Hessian Low Mountain Range Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Heat accumulated above 5 °C over the year - a measure of how much growing season a crop gets.
Mean annual air temperature, averaged over the reference period.
Total precipitation summed over the year.
Precipitation falling in the warmest three-month quarter - relevant to summer drought stress.

Hessian Low Mountain Range

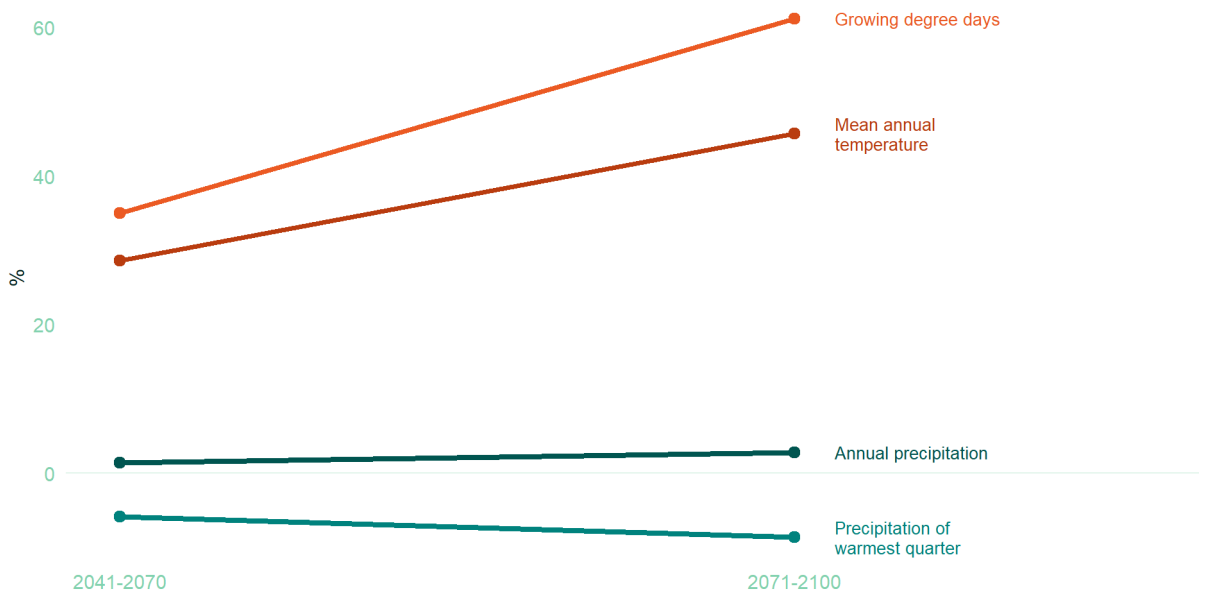


Figure 4: Chart showing Climate in the Hessian Low Mountain Range Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Overview

Where low mountain ranges are aligned with the prevailing wind direction, heavy orographic rainfall is to be expected, whilst precipitation is lower in the lee of the mountains. Temperatures are lower here than on the plains. Snow cover in winter is deeper and lasts longer than in the lowlands.

Challenges

In the low mountain ranges, heavy rainfall events are becoming more frequent and there are a greater number of hot days. This can have a detrimental effect on forage production.

Soil

Key figures

NITROGEN SURPLUS	PHOSPHORUS SURPLUS	GROUNDWATER ABSTRACTION (NON-PUBLIC SUPPLY)
No data available kg N/ha UAA	**No data available** kg P/ha UAA	**2,244** 1000 m3

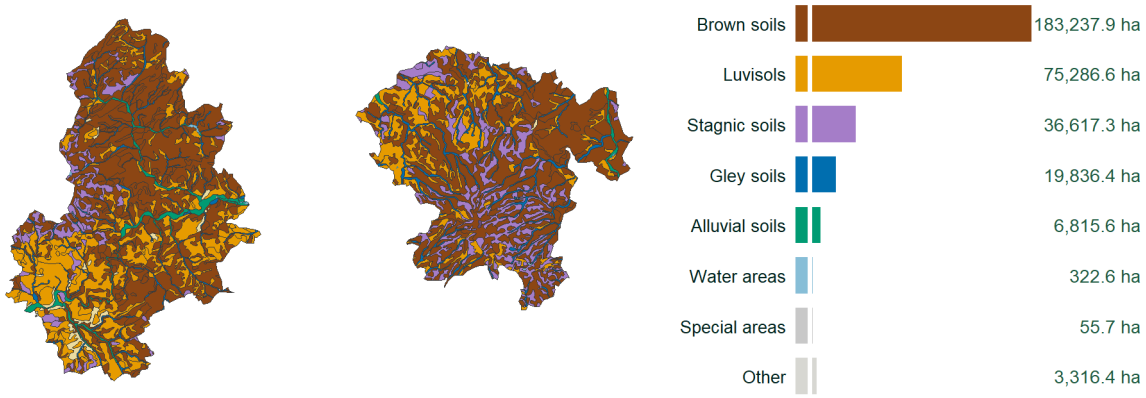


Figure 5: Map showing Soil in the Hessian Low Mountain Range Living Lab (Data: Soil overview map (BUEK250))
Legend shows the dominant semantic soil groups for this Living Lab; raw BUEK IDs stay in the data only as provenance.

Overview

Soil types are highly variable: Podzolic brown soils with low base saturation, red sandstone, brown soils with high to medium base saturation, pseudogley, parabraun soils from loess with high base saturation, rendzina on calcareous sites (locally limited).

Challenges

Soil erosion during heavy rainfall, locally shallow non arable soils

Socio-economic

Key figures

TOTAL POPULATION	GDP PER CAPITA	UNEMPLOYMENT RATE	DISPOSABLE HOUSEHOLD INCOME PER CAPITA
1,041,453 persons	43,153.6 EUR	5.34 %	26,575.6 EUR



Figure 6: Map showing Socio-economic in the Hessian Low Mountain Range Living Lab (Data: Standard land values (BORIS))

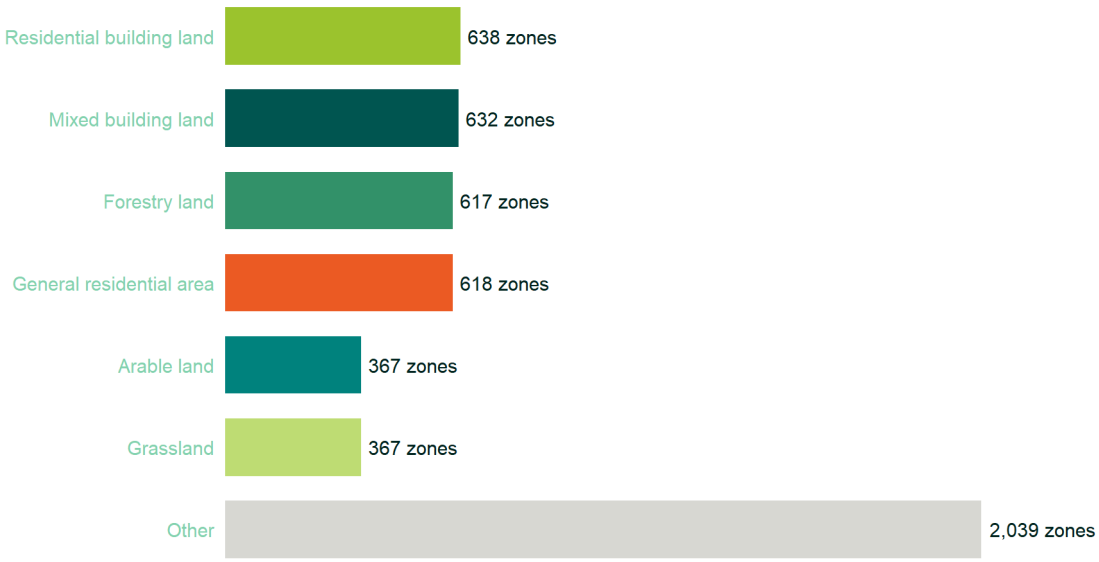


Figure 7: Chart showing Socio-economic in the Hessian Low Mountain Range Living Lab (Data: Standard land values (BORIS))

Overview

Peripheral areas — long distances to regional centres. Population structure: decline, ageing. Comparatively low disposable household income.

Challenges

Sales market for products (milk, meat, etc.) located far away, lower direct marketing opportunities, Long distances to slaughterhouses and processing companies. Shortage of workers due to migration and ageing population.

Landscape

Key figures

FOREST AREA	NATURA 2000 AREA	NATURE RESERVES AREA	SHARE OF SEALED SURFACE
214,513 ha	93,449.7 ha	4,475.8 ha	15 %

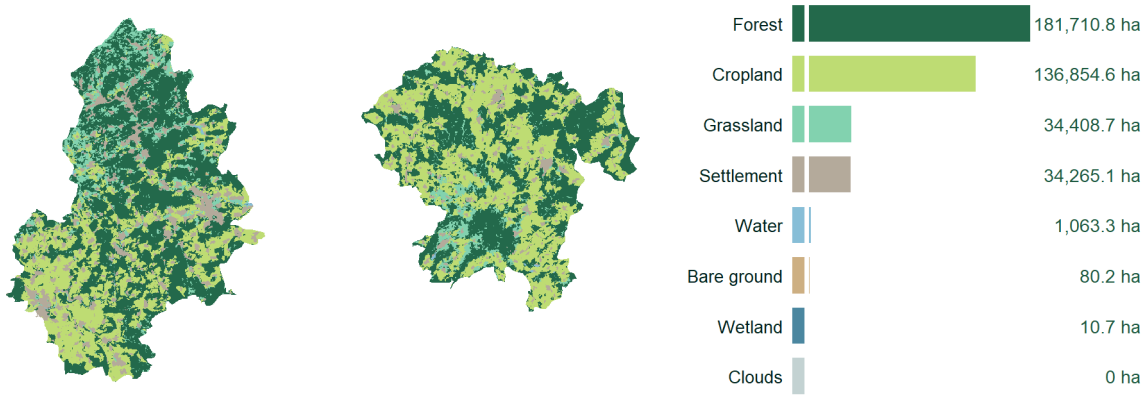


Figure 8: Map showing Landscape in the Hessian Low Mountain Range Living Lab (Data: Land cover (Esri/Impact Observatory, 2024))

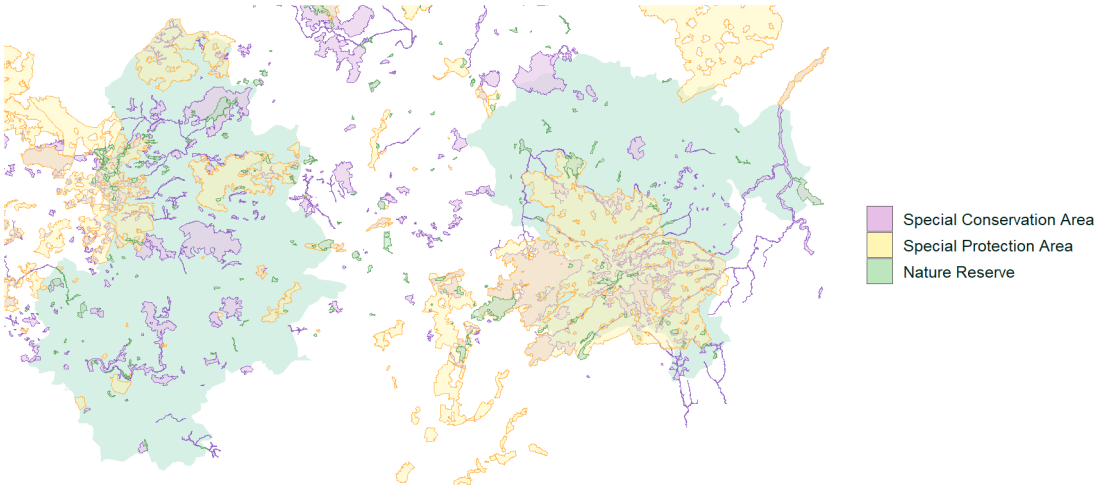


Figure 9: Map showing Protected Areas in the Hessian Low Mountain Range Living Lab (Data: Protected areas (BfN, 2019/2023))

Conservation sites intersecting this Living Lab region. Full polygon boundaries are shown, so sites extend beyond the region outline and designations may overlap. Source: BfN (FFH and bird sanctuaries 2019, nature reserves 2023) - not suitable for planning purposes.

Overview

Low mountain regions 400-1000 m altitude with high relief energy. The agricultural landscape is characterised by hedgerows and field margins and a patchwork of grassland and arable fields. The continuation of grassland farming depends on the maintanance of cattle farms.

Challenges

Peripheral areas are under considerable pressure due to the construction of wind farms and solar power plants. The use of large areas of land is restricted by nature conservation legislation (e.g. Natura 2000). The decline in the number of full-time and, in particular, part-time farmers will result in a change to the landscape, leading to more uniform structures.